

Health Halos, Food Marketing, and Nutritional Quality in Packaged Foods

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Introduction

Food packaging does more than tell consumers what is inside a product. It is also marketing, and the information that stands out most on a package can shape how people understand a food before they ever read the full nutrition label. Claims and cues such as “high protein,” “low fat,” “no added sugar,” “organic,” and “plant-based” draw attention to a positive characteristic, but they do not necessarily describe the nutritional quality of the product as a whole. Research on the health halo effect shows that people can take one favorable feature and form a broader impression that a product is healthy, even when that impression goes beyond what the claim actually says (Roe et al., 1999; Schuldt & Schwarz, 2010). Researchers have therefore asked two related questions: how packaging shapes consumer perception, and whether products carrying these claims actually have stronger nutritional profiles. My project builds on both lines of work by asking a narrower question. How reliably do specific claims and cues signal nutritional quality, and does that relationship change across categories of packaged food?

Health claims and consumer perception

One finding that appears consistently in this literature is that consumers do not always read health information as narrowly as it is written. Roe et al. (1999) found that health and nutrient claims could draw attention toward favorable information on the front of a package and shorten consumers’ search for other nutrition information. Chandon (2013) places this effect within the broader role of packaging, arguing that claims, package design, serving information, and visual cues can all shape perceptions of healthfulness

and consumption. Taken together, these studies suggest that a claim does not have to be false to leave an impression that goes beyond what the claim actually establishes.

More recent work finds the same problem with specific cues. McKeon and Hallman (2024) compared consumer perceptions of Special K Original and Special K Protein. Participants viewed the protein version as healthier and more nutritious, even though, after adjusting for serving size, it contained more sugar, sodium, and calories than the original. Schuldt and Schwarz (2010) found a similar halo from an “organic” label. Participants judged identical cookies as lower in calories when they were described as organic, even though the nutrition information had not changed. I find these two studies useful together because “high protein” refers directly to nutrient content while “organic” refers to how a food was produced, yet both can influence broader judgments about healthfulness.

Claims also compete with other information on the package. Franco-Arellano et al. (2020) found that nutrition labeling systems on the front of packages could help consumers recognize less healthy products, but some health claims still raised perceived healthfulness. Kim et al. (2024) similarly found that health claims and promotional messages on cereal packages worked together to shape how consumers evaluated a product. Both studies suggest that people respond to a package as a combination of signals rather than judging each statement on its own. At the same time, much of this experimental work uses a small number of products, such as cereal, cookies, or beverages, under controlled conditions. There is strong evidence that packaging can change perception, but it tells us less about whether the same claims reliably identify nutritionally stronger products across the wider food market.

How well do claims reflect nutritional quality?

Studies that look at actual products give a more mixed answer. Kaur et al. (2016) sampled 2,034 foods across five European countries and separated nutrition claims from health claims. In their framework, nutrition claims describe beneficial nutritional properties tied to the energy or nutrients a food contains, contains in increased or reduced amounts, or does not contain, while health claims connect a food or one

of its components to health. This distinction matters for my project because “high protein,” “low fat,” and “no added sugar” fit most naturally as nutrition claims, whereas “organic” and “plant-based” do not fit neatly into that regulatory category. Kaur et al. found that 61% of foods carrying nutrition claims passed the nutrient profile model used in the study, compared with 36% of foods carrying neither type of claim. Among foods carrying health claims, 70% passed. Even so, the authors described the overall nutritional differences as relatively modest, which suggests that claims can carry some useful information without being a dependable measure of the whole product.

Breen et al. (2020) found weaker evidence that marketing language signals better nutrition. In a study of more than 2,000 snack products, foods marketed as health foods carried more claims and marketing buzzwords, but the number of claims did not consistently line up with a stronger nutrient profile. Read together, Kaur et al. and Breen et al. do not support a simple conclusion that claimed products are either clearly healthier or clearly no different. The two studies may also differ because of how they sampled products, which food categories they covered, how they defined claims, and which nutrient profile models they used. To me, this disagreement is a reason to look more closely at what type of claim is being made and what type of product carries it.

Kaur et al.’s study is especially important for defining what my project adds, because it already examined broad differences between nutrition and health claims and took food categories into account. My project therefore has to be more specific. Rather than treating nutrition claims or health claims as broad groups, I plan to separate individual cues such as high protein, low fat, no added sugar, organic, and plant-based. I will then test whether each cue is associated with nutritional quality and whether that relationship changes across breakfast cereals, yogurts, snack and protein bars, and packaged snacks. This shifts the question from “Are foods with claims healthier?” to “Which claims are stronger or weaker signals of nutrition in particular product categories?”

Skills and methods for measuring nutritional quality

Answering that question requires a clear way to measure nutritional quality. My primary outcome will be Nutri-Score, which places foods into five ordered categories from A (best nutritional rating) through E. It is useful because it gives a standardized overall measure, but it should not be treated as an objective definition of whether a food is healthy. Van der Bend et al. (2022) note that there is no accepted gold standard for evaluating the healthfulness of individual foods and that the validity and applicability of Nutri-Score call for cautious interpretation. The system was also developed and validated largely in a European context. Because of these limits, I will also examine individual measures such as sugar, sodium, saturated fat, fiber, and protein, so that my conclusions do not depend entirely on a single nutrient profile system.

The data source brings its own set of skills and limitations. I plan to use Open Food Facts, a collaborative database of product labels, ingredients, categories, and nutrition information. Chazelas et al. (2020) showed that Open Food Facts can support research at this scale by analyzing 126,556 products from the French market. Because the database relies on crowdsourced product information, using it also requires careful cleaning and quality checks. I will need to standardize claim labels and food categories, examine missing nutrition values, and decide which records have enough information to include in the analysis.

The statistical methods also need to match the structure of the data. Kaur et al. (2016) used comparisons and regression models that accounted for food category, which shows why category cannot be ignored when both claim prevalence and nutrition vary across food groups. Nutri-Score is ordered from A through E, so I plan to use ordinal logistic regression instead of treating the categories as equally spaced numbers (McCullagh, 1980). A proportional odds model assumes that a predictor has a consistent relationship across the cumulative divisions of the ordered outcome, so I will check that assumption rather than take it for granted (Brant, 1990). I will also include interactions between claim type and food category to test whether, for example, a claim about high protein has the same relationship with nutritional quality in yogurt as it does in snack bars.

Conclusion and broader importance

Overall, the literature shows that packaging that emphasizes health can influence judgments beyond the literal meaning of a claim, while research on actual products gives a less consistent answer about whether those claims point to nutritionally stronger foods. That makes the relationship worth studying, both analytically and for consumers. People often see claims on the front of a package before they look at detailed nutrition information, so a claim that consistently lines up with a stronger nutritional profile could be genuinely useful, while a weaker signal could encourage an overly positive impression.

My project builds on this literature by examining finer claim categories, several packaged food categories, Nutri-Score, and individual nutrients within the same analysis. Because I am not collecting consumer data, it will not show that a particular claim causes consumers to make a certain decision. Instead, it will test how well the positive signals emphasized on packages line up with the nutritional characteristics behind them. Knowing which claims are stronger or weaker signals could help consumers interpret food marketing more carefully and contribute to broader discussions about how health information is communicated on packaged foods.

AI Use Statement

Generative AI (ChatGPT; OpenAI, 2026) was used during the development of this literature review to brainstorm research directions and identify some potentially relevant scholarly references. The sources, claims, and interpretations included in the final paper were written and reviewed by me.

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